FLUCTUATION ASYMPTOTICS OF THE PARABOLIC ANDERSON MODEL

PH.D. CONFIRMATION REPORT

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INTRODUCTION

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This confirmation addresses the fluctuation asymptotics for the Parabolic Anderson Model (PAM) in non-Euclidean geometric settings and in singular regimes.

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The first part shows the progress for the PAM on the Heisenberg group driven by a centred left-stationary Gaussian potential.

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Assuming some regularity on the covariance, we establish second-order (fluctuation) asymptotics for fixed moments of the solution.

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The resulting fluctuation constant is characterised by a variational problem combining a Donsker-Varadhan functional with an explicit limit functional obtained from the second-order expansion of the covariance under Heisenberg (anisotropic) dilations.

INTRODUCTION

The second part outlines ongoing work on the singular PAM on the hyperbolic plane driven by spatial white noise, where well-posedness requires renormalisation, analytic controls of the heat kernel and weighted spaces of distributions.

INTRODUCTION

The final part describes the localisation phenomena beyond the lattice setting. As a next step in the research, my goal is to identify analogues in the regular tree and the continuum.

THE PAM IN THE HEISENBERG GROUP

PROBLEM STATEMENT AND MAIN RESULT

The Parabolic Anderson Model in the Heisenberg group $\mathbf{H}$ is given by

$$\begin{cases} \partial_t u(t,g) = \Delta u(t,g) + \xi(g)u(t,g), & t \in (0,\infty),\ g \in \mathbf{H} \\ u(0,g) = 1, & g \in \mathbf{H} \end{cases} \tag{1}$$

where Δ is the sub-Laplacian, and ξ is a centred Gaussian field that is left-stationary under the action of $\mathbf{H}$.

Denoting the generators of the Heisenberg algebra $\mathfrak{h}$ by $X = \partial_x - \frac{y}{2}\partial_z$, $Y = \partial_y + \frac{x}{2}\partial_z$ and $Z = \partial_z$, the sub-Laplacian above is

$$\Delta = -(X^2 + Y^2).$$

LOG MGF

We introduce **cumulant generating function**

$$\mathrm{H}(t) = \log\langle e^{t\xi(0)}\rangle = \frac{t^2\sigma^2}{2} < \infty, \quad \forall\, t > 0$$

where $\sigma^2 \triangleq \mathrm{C}(0)$, and C is the covariance function.
Throughout the text, we denote the expectation and probability with respect to ξ by $\langle\cdot\rangle$ and $\mathrm{Prob}(\cdot)$, respectively. We'll use g, h to denote elements in the Heisenberg group **H** and write their coordinates as $g = (x, y, z)$.

ANISOTROPIC DILATION AND RENORMALISED POTENTIAL

Recall that, for $r > 0$, the **anisotropic dilation** δ_r on the Heisenberg group is defined as

$$\delta_r(x, y, z) = (rx, ry, r^2 z), \quad g = (x, y, z) \in \mathbf{H}$$

Using the dilation, we define the **normalised potential** as

$$\xi_t(g) = \alpha^2(t) \left(\xi(\delta_{\alpha(t)} g) - \frac{\mathrm{H}(t)}{t} \right)$$

where $\alpha(t)$ is a positive **scale function** which tends to zero as $t \to \infty$. We also define $\beta(t) \triangleq t/\alpha^2(t)$, which is the **large deviation scale**.

RESCALED CUMULANT

Let $\mathcal{P}_c(\mathbf{H})$ denote the set of compactly supported probability measures on $\mathbf{H}$. We will assume that the **rescaled cumulant** $\mathrm{J}_t : \mathcal{P}_c(\mathbf{H}) \longrightarrow [0, \infty)$, defined by

$$\mathrm{J}_t(\mu) = \frac{-1}{\beta(t)} \log\langle e^{\beta(t)\langle \mu, \xi_t \rangle} \rangle, \tag{2}$$

will converge, as $t \to \infty$, to a functional $\mathrm{J} : \mathcal{P}_c(\mathbf{H}) \longrightarrow [0, \infty)$ uniformly on $\mathcal{P}(\mathrm{K})$.

MAIN RESULT

Theorem 1. *Given $p \in [1, \infty)$ fixed, we have the following second order asymptotics for the solution $u(t, g)$ of the PAM* (1)*:*

$$\lim_{t \to \infty} \frac{1}{\beta(pt)} \log(e^{-\mathrm{H}(pt)} \langle u(t, 0)^p \rangle) = -\chi \tag{3}$$

where χ is the fluctuation constant.

In other words, we have

$$\lim_{t \to \infty} \langle u(t, 0)^p \rangle = \exp[\mathrm{H}(pt) - \beta(pt)(\chi + o(1))] \tag{4}$$

AUXILIARY RESULTS

BOXED PROBLEM

Let $r > 0$ and define Q_r to be the ball of radius r with respect to a fixed homogeneous norm. We introduce the following initial value problem

$$\begin{cases} \partial_t u_r^{\mathrm{V}}(t,g) = \Delta u_r^{\mathrm{V}}(t,g) + \mathrm{V}(g) u_r^{\mathrm{V}}(t,g), & t \in (0,\infty),\ g \in \mathrm{Q}_r \\ u_r^{\mathrm{V}}(0,g) = 1, & g \in \mathrm{Q}_r \\ u_r^{\mathrm{V}}(t,g) = 0, & t \in (0,\infty),\ g \in \partial \mathrm{Q}_r \end{cases} \tag{5}$$

with $\mathrm{V} : \mathbf{H} \longrightarrow \mathbf{R}$ an arbitrary Hölder continuous potential.

SCALING RELATIONS

The Brownian motion in the (first) Heisenberg group is given by $W_t = (B_t, S_t)$, where $B_t = (B_t^1, B_t^2)$ is the standard two-dimensional Brownian motion, and S_t is the Lévy area:

$$S_t = \frac{1}{2}\int_0^t (B_s^1\, dB_s^2 - B_s^2\, dB_s^1)$$

Notice that the Brownian scaling becomes

$$(B^1(t), B^2(t), S(t)) = (\sqrt{c}B^1(t/c), \sqrt{c}B^2(t/c), cS(t/c)) \quad \text{in law}$$

Lemma 2 (Scaling relation). *For every* $R > 0$ *and* $t > 0$*,*

$$u^{\xi}_{R\alpha(t)}(s, g) = e^{sH(t)/t} u^{\xi_t}_{R}(\alpha^{-2}(t)s, \delta_{\alpha^{-1}(t)}g), \quad (s, g) \in (0, \infty) \times Q_{R\alpha(t)}$$

EIGENVALUE SCALING

Let $\lambda_1^V > \lambda_2^V \geq \lambda_3^V \geq \cdots$ be the eigenvalues of $\Delta + V$ in $L^2(Q_r)$ with zero boundary condition.

Lemma 3 (Eigenvalue scaling). *For every* $R > 0$, $t > 0$ *and* $k \in \mathbf{N}$, *one has*

$$t\lambda_k^{\xi}(Q_{R\alpha(t)}) = \beta(t)\lambda_k^{\xi_t}(Q_R) + H(t)$$

FEYNMAN-KAC REPRESENTATIONS

Let $p_r^{\mathsf{V}}(t, \cdot, h)$ denote the fundamental solution of the equation (5), but with initial condition

$$p_r^{\mathsf{V}}(0, \cdot, h) = \delta_h(\cdot), \quad h \in \mathsf{Q}_r$$

Recall the normalised occupational measure of Brownian motion

$$\mathsf{L}_t(\mathrm{d}g) = \frac{1}{t} \int_0^t \mathbf{1}\{\mathsf{W}_u \in \mathrm{d}g\} \, \mathrm{d}u$$

with $\{\mathsf{W}_t\}$ a Brownian motion in $\mathbf{H}$ with generator Δ. We denote by $\mathbf{E}_g$ and $\mathbf{P}_g$ the underlying expectation and probability when $\mathsf{W}_0 = g \in \mathbf{H}$, respectively.

FEYNMAN-KAC REPRESENTATIONS

Letting $\tau_r = \inf\{t \geq 0 : \mathsf{W}_t \notin \mathsf{Q}_r\}$ be the exit time from Q_r, we have the Feynman-Kac representations

$$u_r^{\mathsf{V}}(t, g) = \mathbf{E}_g \exp\left[\int_0^t \mathsf{V}(\mathsf{W}_u)\, \mathrm{d}u\right] \mathbf{1}\{\tau_r > t\} = \mathbf{E}_g \exp[t\langle \mathsf{L}_t, \mathsf{V}\rangle]\mathbf{1}\{\operatorname{supp} \mathsf{L}_t \subset \mathsf{Q}_r\} \tag{6}$$

and

$$p_r^{\mathsf{V}}(t, g, h) = \mathbf{E}_g \exp[t\langle \mathsf{L}_t, \mathsf{V}\rangle]\mathbf{1}\{\operatorname{supp} \mathsf{L}_t \subset \mathsf{Q}_r\}\delta_h(\mathsf{W}_t) \tag{7}$$

SPECTRAL REPRESENTATION

Letting (e_k) be an orthonormal basis of $L^2(Q_r)$, we have the spectral decompositions

$$p_r^{\vee}(t, g, h) = \sum_{k=1}^{\infty} \exp(t\lambda_k) e_k(g) e_k(h) \tag{8}$$

and

$$u_r^{\vee}(t, \cdot) = \sum_{k=1}^{\infty} \exp(t\lambda_k)(e_k, \mathbf{1})_r e_k(\cdot) \tag{9}$$

where $(\cdot, \cdot)_r$ denotes the inner product on $L^2(Q_r)$.
It is easy to check that the corresponding semigroup is ultracontractive in the sense of Davies [Dav89]. The spectral decomposition then follows from [Dav89, Theorem 2.1.4].

PROOF OF THE LOWER BOUND

LB: RESTRICTION TO A BALL

In this section, we prove the following lower bound

$$\lim_{t\to\infty} \langle u(t,0)^p \rangle \geq \exp[\mathrm{H}(pt) - \beta(pt)(\chi + o(1))] \tag{10}$$

The first step is to show that we can restrict our solution to a ball. To prove that, we need to verify that the solution is spatially invariant in the following sense.

Lemma 4. *For any $g \in \mathbf{H}$, we have $u(t,0) = u(t,g)$ in law.*

LB: RESTRICTION TO A BALL

Lemma 5 (Restriction to a ball). *For every* $\mathrm{R} > 0$ *and any* $p \geq 1$,

$$\langle u^{\xi}(t,0)^{p} \rangle \geq \langle (u^{\xi}_{\mathrm{R}\alpha(pt)}(t,\cdot), \mathbf{1})^{p} \rangle$$

LB: REDUCTION TO FIRST MOMENT

Lemma 6 (Reduces to the case $p = 1$***).*** *For every* $\mathrm{R} > 0$ *and any* $p \geq 1$,

$$\langle (u^{\xi}_{\mathrm{R}\alpha(pt)}(t,\cdot), \mathbf{1})^p \rangle \geq e^{o(\beta(pt))} \langle (u^{\xi}_{\mathrm{R}\alpha(pt)}(pt,\cdot), \mathbf{1}) \rangle \quad \text{as } t \to \infty \tag{11}$$

Proof. This proof uses the spectral representation (9) and follows by the same argument as in the Euclidean case [GK00]. □

LB: PROOF FOR THE FIRST MOMENT

Lemma 7 (Lower bound for** $p = 1$**). *There exists a constant* $\chi_{\mathsf{R}} > 0$ *such that, as* $t \to \infty$*,*

$$\langle (u^{\xi}_{\mathsf{R}\alpha(t)}(t, \cdot), \mathbf{1}) \rangle = \exp[\mathsf{H}(t) - \beta(t)(\chi_{\mathsf{R}} + o(1))]$$

Furthermore, $\chi_{\mathsf{R}} \to \chi$ *as* $\mathsf{R} \to \infty$*.*

Proof. Applying the scaling relation (Lemma 2) and using Varadhan's lemma [Dem09, Theorem 4.3.1] as well as the fact that the family of measures $\{\mathrm{Law}(\mathsf{L}_{\beta(t)}) : t > 0\}$ satisfies a large deviation principle with rate function $\mathcal{S}_d$ [Dem09, Section 6.5.1], the proof follows as in the Euclidean case [GK00, Lemma 2.(i)]. □

Combining the steps above, we obtain the lower bound (10).

PROOF OF THE UPPER BOUND

HEISENBERG HEAT KERNEL ESTIMATE

By [BOTW23, Equation 2.14], we have the following bounds on the heat kernel.

Lemma 8. *There exist positive constants* $\mathsf{C}_1, \mathsf{C}_2, \mathsf{C}_3$ *and* C_4 *such that for every* $t > 0$ *and* $g \in \mathbf{H}$ *we have*

$$\mathsf{C}_1 t^{-2} \exp\left(-\frac{\mathsf{C}_2}{t} d_{cc}(0,g)^2\right) \leq p_t(g) \leq \mathsf{C}_3 t^{-2} \exp\left(-\frac{\mathsf{C}_4}{t} d_{cc}(0,g)^2\right)$$

Here, d_{cc} *denotes the Carnot-Carathéodory distance.*

HEISENBERG HEAT KERNEL ESTIMATE

Corollary 9 (Heat kernel estimate). *There exists a constant* $\mathsf{C} > 0$ *such that*

$$p_\delta(h, g) \leq \mathsf{C}\delta^{-2}$$

for every $\delta > 0$ *and* $h, g \in \mathbf{H}$.

We remark that the kernel estimate above is equivalent to a Sobolev inequality and a volume growth condition. See [VSCC93].

FIXED-DOMAIN UPPER BOUND

Lemma 10 (Fixed-domain upper bound). *There exists a constant* $\chi_{\mathsf{R}} > 0$ *such that, as* $t \to \infty$*,*

$$\left\langle \sum_{k=1}^{\infty} \exp(t\lambda_k^{\xi}(\mathsf{I}_t)) \right\rangle \leq \exp[\mathsf{H}(t) - \beta(t)(\chi_{\mathsf{R}} + o(1))]$$

Furthermore, $\chi_{\mathsf{R}} \to \chi$ *as* $\mathsf{R} \to \infty$*.*

Proof. The proof is exactly as in the Euclidean case [GK00, Lemma 2(ii)], using the eigenvalue scaling (Lemma 3) and the spectral decomposition (8). □

DECOMPOSITION OF THE PRINCIPAL EIGENVALUE

Given a potential V and a bounded domain D, we want to estimate the principal Dirichlet eigenvalue on D, with a sacrifice on the potential, in terms of the principal eigenvalue on smaller domains D_m.

DECOMPOSITION OF THE PRINCIPAL EIGENVALUE

Let $\{\varphi_m : m \in \mathbf{N}\}$ be a smooth partition of unity with $\sum_m \varphi_m^2 = 1$ and set $\mathsf{D}_m \triangleq \operatorname{supp} \varphi_m$.

Lemma 11. *Let* $\Phi(x) \triangleq \sum_m |\nabla \varphi_m|^2$. *Then one has*

$$\lambda^{\mathsf{V}-\Phi}(\mathsf{D}) \leq \sup_m \lambda^{\mathsf{V}}(\mathsf{D}_m)$$

where ∇ *is the* ***horizontal gradient*** $\nabla f \triangleq (\mathsf{X}f, \mathsf{Y}f)$.

Proof. Let $\psi \in \mathsf{C}_c^\infty(\mathsf{D})$, and also define $\psi_m \triangleq \psi\varphi_m \in \mathsf{C}_c^\infty(\mathsf{D}_m)$. Using that $\sum_m \varphi_m^2 = 1$, the result follows from the Rayleigh-Ritz formula. $\square$

DECOMPOSITION OF THE PRINCIPAL EIGENVALUE

To guarantee that the sacrifice of potential is small, we need a partition of unity that respects the decomposition of our domain. We decompose the large ball $\mathrm{Q_R}$ by using sphere packing.

Let $\mathcal{C}(r) = \{\mathrm{B}(g_i, r/2) : g_i \in \mathrm{Q_R},\ i \in \mathcal{I}\}$ denote a maximal sphere packing of $\mathrm{Q_R}$. Then, the family $\mathcal{U}(r) \triangleq \{\mathrm{B}(g_i, r) : g_i \in \mathrm{Q_R},\ i \in \mathcal{I}\}$ is an open cover of $\mathrm{Q_R}$.

DECOMPOSITION OF THE PRINCIPAL EIGENVALUE

Lemma 12. *For each $r > 1$, there exists a partition of unity*

$$\{\varphi_i : i \in \mathcal{J}\}$$

on $\mathrm{Q}_{\mathrm{R}(t)}$ subordinate to the open cover $\mathcal{U}(r)$ defined above, i.e., the support supp φ_k is contained in a ball $\mathrm{B}(g_k, r)$ for every k.
Furthermore, we have the gradient bound $\Phi_r \leq \mathrm{C}/r^2$ for some universal constant C.

TRUNCATING THE DOMAIN

Our goal in this section is to localise the solution in a ball or radius $R(pt)$. In order to do that, we first need the following estimate.

Lemma 13 (Exit time estimate). *There exist positive constants* C_1 *and* C_2 *such that*

$$\mathbf{P}_0(\tau_R \leq t) \leq C_1 e^{-C_2 R^2/t} \tag{12}$$

for all $R \gg t > 1$.

TRUNCATING THE DOMAIN

Lemma 14 (Localisation). *As $t \to \infty$, we have that*

$$\langle u^{\xi}(t,0)^{p} \rangle \leq e^{o(\beta(t))} \langle u^{\xi}_{\mathrm{R}(pt)}(t,0)^{p} \rangle$$

Proof. Using the exit time estimate (12) and the lower bound (10), the proof is the same as Lemma 3 in [GK00]. □

REDUCTION TO THE CASE $p = 1$

Lemma 15. *As $t \to \infty$, we have that*

$$\left\langle u^{\xi}_{\mathrm{R}(pt)}(t,0)^p \right\rangle \leq e^{o(\beta(t))} \left[\left\langle (u^{\xi}_{\mathrm{R}(pt)}(pt,\cdot), \mathbf{1}) \right\rangle + e^{\mathrm{H}(pt) - \beta(pt)\chi} \right]$$

Proof. See [GK00, Lemma 4]. □

PROOF OF THE UPPER BOUND FOR $p = 1$

Lemma 16. *As $t \to \infty$,*

$$\left\langle (u^{\xi}_{\mathsf{R}(t)}(t, \cdot), \mathbf{1}) \right\rangle \leq \exp[\mathsf{H}(t) - \beta(t)(\chi + o(1))]$$

PROOF OF THE UPPER BOUND FOR $p = 1$ [1]

Proof. **Step 1. Scaling.** Applying (2),

$$\langle (u^{\xi}_{\mathsf{R}(t)}(t,\cdot), \mathbf{1})\rangle e^{-\mathsf{H}(t)} = e^{o(\beta(t))} \langle (u^{\xi_t}_{\tilde{\mathsf{R}}(t)}(t,\cdot), \mathbf{1})\rangle$$

with $\tilde{\mathsf{R}}(t) \triangleq \mathsf{R}(t)/\alpha(t)$. Thus,

$$\varlimsup_{t\to\infty} \frac{1}{\beta(t)} \log\langle e^{-\mathsf{H}(t)} u_{\mathsf{R}(t)}(t,0)\rangle \le \varlimsup_{t\to\infty} \frac{1}{\beta(t)} \log\langle (u^{\xi_t}_{\tilde{\mathsf{R}}(t)}(\beta(t),\cdot), \mathbf{1})\rangle$$

PROOF OF THE UPPER BOUND FOR $p = 1$ [2]

Step 2. Sacrifice of potential. Since $\Phi_r \triangleq \sum_k |\nabla \varphi_k|^2 \leq \mathrm{C}/r^2$, using Feynman-Kac we have

$$
\begin{aligned}
u^{\xi_t}_{\tilde{\mathrm{R}}(t)}(\beta(t), x) &= \mathbf{E}_x \left[\exp\left(\int_0^{\beta(t)} \xi_t(\mathrm{W}_s)\, \mathrm{d}s \right) ; \mathrm{W}([0, \beta(t)]) \subseteq \mathrm{Q}_{\tilde{\mathrm{R}}(t)} \right] \\
&\leq e^{\beta(t)\mathrm{C}/r^2} \mathbf{E}_x \left[\exp\left(\int_0^{\beta(t)} (\xi_t - \Phi_r)(\mathrm{W}_s)\, \mathrm{d}s \right) ; \mathrm{W}([0, \beta(t)]) \subseteq \mathrm{Q}_{\tilde{\mathrm{R}}(t)} \right] \\
&= e^{\mathrm{C}\beta(t)/r^2} u^{\xi_t - \Phi_r}_{\tilde{\mathrm{R}}(t)}(\beta(t), x)
\end{aligned}
$$

Hence,

$$
(u^{\xi_t}_{\tilde{\mathrm{R}}(t)}(\beta(t), \cdot), \mathbf{1}) \leq e^{\mathrm{C}\beta(t)/r^2} (u^{\xi_t - \Phi_r}_{\tilde{\mathrm{R}}(t)}(\beta(t), \cdot), \mathbf{1})
$$

PROOF OF THE UPPER BOUND FOR $p = 1$ [3]

Step 3. Dominance by the principal eigenvalue. We claim that

$$\langle (u^{\xi_t - \Phi_r}_{\widetilde{\mathsf{R}}(t)}(\beta(t), \cdot), \mathbf{1}) \rangle \leq \mathrm{vol}(\mathsf{Q}_{\widetilde{\mathsf{R}}(t)}) \langle e^{\beta(t)\lambda^{\xi_t - \Phi_r}} \rangle$$

Indeed, using the spectral representation (9), we have

$$u^{\xi_t - \Phi_r}_{\widetilde{\mathsf{R}}(t)}(\beta(t), \cdot) = \sum_{k=1}^{\infty} e^{\beta(t)\lambda_k^{\xi_t - \Phi_r}} (e_k, \mathbf{1}) e_k(\cdot) \leq e^{\beta(t)\lambda^{\xi_t - \Phi_r}} \sum_{k=1}^{\infty} (e_k, \mathbf{1}) e_k(\cdot)$$

Thus,

$$(u^{\xi_t - \Phi_r}_{\widetilde{\mathsf{R}}(t)}(\beta(t), \cdot), \mathbf{1}) \leq e^{\beta(t)\lambda^{\xi_t - \Phi_r}} \|\mathbf{1}\|^2_{\widetilde{\mathsf{R}}(t)} = \mathrm{vol}(\mathsf{Q}_{\widetilde{\mathsf{R}}(t)}) e^{\beta(t)\lambda^{\xi_t - \Phi_r}}$$

Taking expectation yields the claim.

PROOF OF THE UPPER BOUND FOR $p = 1$ [4]

Using that the volume of a ball of radius r in the Heisenberg group is of order r^4, we conclude that

$$\overline{\lim_{t\to\infty}} \frac{1}{\beta(t)} \log\langle (u^{\xi_t}_{\widetilde{\mathsf{R}}(t)}(\beta(t),\cdot), \mathbf{1})\rangle \leq \overline{\lim_{t\to\infty}} \frac{1}{\beta(t)} \log\langle e^{\beta(t)\lambda^{\xi_t - \Phi_r}(\mathsf{Q}_{\widetilde{\mathsf{R}}(t)})}\rangle$$

PROOF OF THE UPPER BOUND FOR $p = 1$ [5]

Step 4. Decomposition of the principal eigenvalue. Applying Lemma 11,

$$\begin{aligned}\langle e^{\beta(t)\lambda^{\xi_t - \Phi_r}(\mathrm{Q}_{\widetilde{\mathrm{R}}(t)})}\rangle &\leq \langle \max_k \exp(\beta(t)\lambda^{\xi_t}(\mathrm{D}_k))\rangle \\ &\leq \sum_k \langle \exp(\beta(t)\lambda^{\xi_t}(\mathrm{D}_k))\rangle\end{aligned}$$

Since $\mathrm{D}_k \subseteq \mathrm{B}_{k,r} = \mathrm{B}(g_k, r)$, Rayleigh formula implies that $\lambda^{\xi_t}(\mathrm{D}_k) \leq \lambda^{\xi_t}(\mathrm{B}_{k,r})$. Hence,

$$\overline{\lim_{t\to\infty}} \frac{1}{\beta(t)} \log\langle e^{\beta(t)\lambda^{\xi_t - \Phi_r}(\mathrm{Q}_{\widetilde{\mathrm{R}}(t)})}\rangle \leq \overline{\lim_{t\to\infty}} \frac{1}{\beta(t)} \log \left\langle \sum_k \exp(\beta(t)\lambda_k^{\xi_t}(\mathrm{B}_{k,r})) \right\rangle$$

PROOF OF THE UPPER BOUND FOR $p = 1$ [6]

Step 5. Fixed-domain upper bound. Using 10 and combining the steps, we have

$$\overline{\lim_{t\to\infty}} \frac{1}{\beta(t)} \log\langle e^{-\mathrm{H}(t)} u(t,0)\rangle \leq -\chi_r + \frac{C}{r^2}$$

Taking $r \to \infty$ completes the proof. $\square$

EXPLICIT COMPUTATION OF THE FLUCTUATION CONSTANT

EXPLICIT COMPUTATION OF THE FLUCTUATION CONSTANT

Our goal is to compute the fluctuation constant

$$\chi \triangleq \inf \left\{ \mathcal{S}(\mu) + \mathsf{J}(\mu) : \mu \in \mathcal{P}_c(\mathbf{H}) \right\}$$

where $\mathcal{S}$ is the Donsker-Varadhan functional

$$\mathcal{S}(\mu) \triangleq \begin{cases} \int_{\mathbf{H}} |\nabla_{\mathbf{H}} \phi|^2 \, \mathrm{d}\mu, & \mu \ll \mathrm{d}g,\ \phi \triangleq \sqrt{d\mu/dg} \in \mathsf{W}^{1,2}(\mathbf{H}); \\ +\infty, & \text{otherwise.} \end{cases}$$

COVARIANCE

Let us consider the limit functional J. To compute it explicitly, we need the second-order expansion of the covariance function

$$\mathrm{C}(g,h) \triangleq \mathrm{cov}[\xi(g),\xi(h)] = \langle \xi(g)\xi(h)\rangle, \quad g,h \in \mathbf{H} \tag{13}$$

TAYLOR EXPANSION IN THE HEISENBERG GROUP

In the Heisenberg group, we have the Taylor expansion

$$
\begin{aligned}
f(x, y, z) &= \sum_{k=0}^{\infty} \frac{1}{k!}[(x\mathrm{X} + y\mathrm{Y} + z\mathrm{Z})^k]f(0) \\
&= f(0) + \sum_{i=1}^{3} \mathrm{X}_i x_i f(0) + \frac{1}{2}\sum_{i,j=1}^{3} \mathrm{X}_i \mathrm{X}_j x_i x_j f(0) + \cdots
\end{aligned}
$$

where $(x, y, z) \in \mathbf{H}$ and we used $\mathrm{X}_1 = \mathrm{X}$, $\mathrm{X}_2 = \mathrm{Y}$, and $\mathrm{X}_3 = \mathrm{Z}$ (see Section 3 of [ACC10]).

THE LIMIT FUNCTIONAL J

Computing $\mathsf{J}_t(\mu)$ explicitly, we obtain

$$\mathsf{J}_t(\mu) = \frac{t\alpha^2(t)}{2} \iint [\mathsf{C}(0) - \mathsf{C}(\delta_{\alpha(t)}(g^{-1}h)]\, \mu(\mathrm{d}g)\mu(\mathrm{d}h) \tag{14}$$

Replacing the Taylor expansion into (14), we have

$$\mathsf{J}_t(\mu) = \frac{-t\alpha^4(t)}{2} \iint_{\mathbf{H}^2} \frac{1}{2}\pi(g^{-1}h)\mathsf{C}''(0)\pi(g^{-1}h)^{\mathsf{T}}\, \mu(\mathrm{d}x)\mu(\mathrm{d}y)$$

where $\pi : \mathbf{H} \longrightarrow \mathbf{R}^2$ is the horizontal projection $(x, y, z) \longmapsto (x, y)$ and $\mathsf{C}'' \triangleq \mathsf{HC}$.

THE LIMIT FUNCTIONAL J

Setting $\alpha(t) = t^{-1/4}$ as in the Euclidean case, supposing that C is twice continuously differentiable and $C''(0) = -\Sigma^2$, and taking the limit as $t \to \infty$, we have that $J_t(\mu)$ converges to

$$J(\mu) = \frac{1}{4} \iint_{\mathbf{H}^2} \pi(g^{-1}h)\Sigma^2\pi(g^{-1}h)^{\mathsf{T}} \, \mu(\mathrm{d}g)\mu(\mathrm{d}h) \tag{15}$$

THE FLUCTUATION CONSTANT

Recall that $\chi = \inf\left\{\mathcal{S}(\mu) + \mathrm{J}(\mu) : \mu \in \mathcal{P}_c(\mathbf{H})\right\}$ and we just computed the limit functional J in (15).

Step 1. To compute χ, the first step is to reduce the problem to

$$\chi = \inf_{\substack{\psi \in \mathrm{C}_c^\infty(\mathbf{H}) \\ \|\psi\|_2 = 1}} ((-\Delta_{\mathbf{H}} + p)\psi, \psi)_{\mathrm{L}^2(\mathbf{H})}$$

where $p(g) = |\Gamma \pi_{x,y}(g)|^2$ for a given positive-definite matrix Γ.

THE FLUCTUATION CONSTANT

Step 2. For $\lambda \in \mathbf{R}$, define the **special Hermite operator**

$$\hat{\mathsf{L}}_\lambda \triangleq \Delta_{x,y} - \frac{\lambda^2}{4}(x^2 + y^2) + i\lambda(x\partial_y - y\partial_x)$$

Then one can show that

$$\widehat{(\Delta_{\mathbf{H}} f)} = \hat{\mathsf{L}}_\lambda \hat{f}$$

where the hat $\hat{}$ denotes the Fourier transform in z.

It is also possible to prove that

$$((-\Delta_{\mathbf{H}} + p)f, f)_{\mathsf{L}^2(\mathbf{H})} = \frac{1}{2\pi} \int_{\mathbf{R}} ((-\hat{\mathsf{L}}_\lambda + p)\hat{f}(\cdot, \cdot, \lambda), \hat{f}(\cdot, \cdot, \lambda))_{\mathsf{L}^2(\mathrm{d}x\mathrm{d}y)} \, \mathrm{d}\lambda$$

Step 3. Compute whole-space bottom spectrum of $-\hat{\mathrm{L}}_\lambda + p$ on $\mathbf{R}^2$. This last step was done for $\Gamma = \mathrm{I}$, and I am finishing the computation for the general case.

QUENCHED ASYMPTOTICS

So far we discussed the annealed (moment) asymptotics of the solution. But what happens in the quenched (almost sure) sense? Here, we want to prove a result of the following form.

Quenched Asymptotics. With probability one,

$$\lim_{t\to\infty} \frac{1}{t} \log u(t,0) = h_t - (\chi_{\text{a.s.}} + o(1))\sqrt{h_t}$$

where $h_t = \sqrt{8\sigma^2 \log t}$ and $\chi_{\text{a.s.}}$ differs from χ by a multiplicative factor.

THE FLUCTUATION CONSTANT

I do have some results about this, but since these haven't been discussed yet, they were not added here. I believe that the key lemma to establish the quenched asymptotics is an estimate for the peak of the Gaussian field, i.e., eventually

$$\xi(g) \geq \left(h_{2^n} - 4\gamma\sqrt{h_{2^n}}\right) p(h^{-1}g) \quad \text{a.s.}$$

with $\gamma > 0$ given, $p(g) = 1 - \frac{1}{2\sigma^2}\left|\Sigma\pi_{x,y}(g)\right|^2$, and g in some enlarged ball that depends on a correct choice of ball packing.

SINGULAR PAM IN THE HYPERBOLIC PLANE

INTRODUCTION

In this project, our objective is to study the Parabolic Anderson Model in the hyperbolic space driven by white-noise potential, including the spectral asymptotics, i.e., eigenvalues of the Anderson operator $\Delta + \xi$.
More precisely, the Parabolic Anderson Model in the Hyperbolic plane $\mathbf{H}^2$ with spatial white noise ξ is given by

$$\partial_t u = \Delta u + u \cdot \xi, \quad u(0, x) = 1 \tag{16}$$

where Δ is the Laplace-Beltrami operator.

INTRODUCTION

As before, our goal is to study the long-time behaviour of the solution, which is formally given by

$$u(t,x) = \mathbf{E}_x \left[e^{\int_0^t \xi(\mathrm{B}_s)\mathrm{d}s - \infty t} \right]$$

where the term "∞t" represents a renormalisation constant.

WELL-POSEDNESS

In order to study long-time asymptotics, we first need to show that the equation (16) is well-posed. We believe that, with the correct design of weights and norms, a Picard iteration will work and give us a global solution.

WELL-POSEDNESS

Choosing the weighted norms appropriately, we have obtained that

$$\|\mathcal{M}_{\mathsf{T}}^{\mathsf{Q}} v\|_{1+\varepsilon;e_0,e_1} \lesssim e^{\mathsf{C}_3}\mathsf{T}^{1-\varepsilon-\kappa} \times \|v\|_{1+\varepsilon;e_0,e_1}(\|g\|_{-\kappa;p_{2a}} + \|h\|_{-\kappa;p_a}).$$

which gives a contraction for a short time horizon T.
Here, $\mathcal{M}$ is the integral map associated with our fixed-point problem, Q is the singular part of the heat kernel, and $\|\cdot\|_{\alpha,w}$ represents a w-weighted α-Hölder norm.

WELL-POSEDNESS

Currently, we are developing estimates for the heat kernel, for the integral transform and for the product between a function and a distribution. For the latter estimate, we are using the Reconstruction Theorem as it was proved by [CZ20] and, for manifolds, by [RS21].

COMPLETE LOCALISATION AND THE TWO-CITIES THEOREM

WHAT IS KNOWN: THE LATTICE CASE

Let us consider now a third version of the Parabolic Anderson Model:

$$\begin{cases} \partial_t u(t,z) = \Delta u(t,z) + \xi(z) u(t,z), & t \in (0,\infty),\ z \in \mathbf{Z}^d \\ u(0,z) = \mathbf{1}_0(z), & z \in \mathbf{Z}^d \end{cases} \tag{17}$$

where Δ is the discrete Laplacian and the potential ξ is a collection on i.i.d. random variables.

WHAT IS KNOWN: THE LATTICE CASE

We call the sum over all $z \in \mathbf{Z}^d$ the **total mass** which we will denote by $\mathrm{U}(t)$. That is,

$$\mathrm{U}(t) = \sum_{z \in \mathbf{Z}^d} u(t, z)$$

WHAT IS KNOWN: THE LATTICE CASE

Suppose that our potential is Pareto distributed, i.e.,

$$\text{Prob}(\xi(z) > x) = x^{-\alpha}, \quad x \geq 1$$

where $\alpha > d$ is the parameter of the distribution.

WHAT IS KNOWN: THE LATTICE CASE

In 2009, [KLMS09] proved the following.

Theorem 17 (Two-Cities Theorem). *There exist two* $\mathbf{Z}^d$*-valued stochastic processes* $Z_t^{(1)} \neq Z_t^{(2)}$ *such that*

$$\lim_{t\to\infty} \frac{u(t, Z_t^{(1)}) + u(t, Z_t^{(2)})}{U(t)} = 1 \quad a.s.$$

Furthermore, the two cities can be chosen such that

$$(Z_t^{(1)}, Z_t^{(2)}) \left(\frac{\log t}{t}\right)^{q+1} \xrightarrow{d} (X^{(1)}, X^{(2)})$$

where $q \triangleq \frac{d}{\alpha - d}$ *and* $(X^{(1)}, X^{(2)})$ *is an explicit distribution.*

WHAT IS KNOWN: THE LATTICE CASE

In the same paper, it was also proved the following.

Theorem 18 (One-point Localisation). *One may choose* $Z_t^{(1)}$ *such that*

$$\lim_{t\to\infty} \frac{u(t, Z_t^{(1)})}{U(t)} = 1$$

in probability.

THE QUESTION: THE TREE AND THE CONTINUUM CASES

Now what happens when we change the space?

And, more importantly, what happens in the continuous case (e.g., the Euclidean space or the hyperbolic space)?

In the hyperbolic space, we expect Sullivan-type asymptotics at the boundary at infinity.

THE QUESTION: THE TREE AND THE CONTINUUM CASES

Before moving into the continuum, we consider the space to be a regular tree with d generators, which we denote by $\mathbf{T}^d$.

In [KLMS09], it was shown that the two cities are given by the first two maxima of

$$\Phi_t(z) \triangleq \left[\xi(z) - \frac{|z|}{t}\log\xi(z) + \frac{\eta(z)}{t}\right]\mathbf{1}\{t\xi(z) \geq |z|\}$$

where $\eta(z) = \log \mathrm{N}(z)$, and $\mathrm{N}(z)$ denotes the number of paths of length $|z|$ starting at the origin and ending at z. Notice that, in the tree, $\mathrm{N} \equiv 1$, so $\eta \equiv 0$.

THE QUESTION: THE TREE AND THE CONTINUUM CASES

We proved that, for $z \in \mathbf{T}^d$, the following lower bound holds

$$\frac{1}{t} \log \mathrm{U}(t) \geq \xi(z) - \frac{|z|}{t} \log \xi(z) + o(1)$$

This gives us the expected version of the function Φ above in the tree setting (just taking $\eta \equiv 0$).

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